

LESSON 10.1

EXPLORING INEQUALITIES – PART 1



LESSON INTRODUCTION

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

LEARNING TARGETS

- I can relate the solution set of an inequality to its graph on a number line.
- I can relate the solution set of an inequality to the context it represents.
- I can determine the minimum or maximum value that makes an inequality true.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

WORKING TOWARD

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

ESSENTIAL QUESTIONS

- What is the minimum or maximum value that makes the inequality true?
- How do you relate the solution set of an inequality to the context it represents?
- How do you determine if solutions are reasonable for the context?

LESSON NARRATIVE

In this lesson, students read, graph, and interpret inequalities in the form $x + p < q$. This lesson is an exploration with scaffolding where students connect algebraic representations of an inequality to a number line and a real-world context. Students explore the meaning of a minimum and maximum value in terms of inequalities and connect these to a real-world context.

COMPONENT SUMMARY

10.1.1 WARM-UP (~5 MIN.)

Identify and justify which inequality does not belong

10.1.2 REFRESHER (5 MIN.)

Represent inequalities on number lines

10.1.3 EXPLORATION (15 MIN.)

Represent inequalities in the context of real-world problems

10.1.4 BRING IT TOGETHER (10 MIN.)

Consider constraints on the solution sets based on the context of inequalities

10.1.5 COLLABORATION (15 MIN.)

Match inequalities and number lines and determine the maximum or minimum values

10.1.6 WRAP-UP (~5 MIN.)

Reflect on mastery of learning targets by completing a Know, Wonder, and Learned table

RESOURCES

GLOSSARY

Key Terms:

- Solution set
- Constraint

Additional Terms:

- Maximum
- Minimum

MATERIALS

- (Optional) Number line templates

ADDITIONAL PREPARATION

- Consider preparing sentence stems for constructive response problems.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle understanding that $x > a$ and $a < x$ are equivalent, because they are not accustomed to the variable being on the right side.
- Students may struggle determining whether to shade the circle on the number line.
- Students may struggle finding the maximum or minimum.
- Students may struggle understanding that the value (p) for a strict inequality ($x > p$ or $x < p$) is not the minimum or maximum.
- Students may have a hard time interpreting the solution sets within the context.

THINGS TO CONSIDER

- Consider creating and displaying an anchor chart and word wall for students to reference throughout the lesson. Include instructions, definitions, examples, and visual models.
- Consider having students practice reading the inequalities aloud from right to left and left to right.
- If time is an issue, consider assigning questions 3-6 in 10.1.5 Collaboration for homework.
- Determine if your students need the support of all or part of the 10.1.2 Refresher.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 10.1.3 Exploration, have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 10.1.4 Bring It Together, students solve, graph, and interpret inequalities based on real-world situations. Students learn that sometimes solutions do not make sense in the context of a problem. For example, if determining the possible height of a person, negative solutions do not make sense.

DIFFERENTIATE INSTRUCTION

In 10.1.5 Collaboration, consider providing students with a collection of numbers to substitute in the inequalities with Guess and Check. This will help students match the inequalities with the number lines and determine the maximum or minimum values.

10.1.1 WARM-UP: WHICH ONE DOESN'T BELONG?

DIFFERENTIATE INSTRUCTION

Students may need a refresher on how to read inequalities. Model how to read inequalities from left to right and from right to left. Then have students practice reading the inequalities themselves.

QUESTIONING

After initially identifying one inequality that does not belong, challenge students to think of as many reasons as they can for each inequality not belonging.



10.1.1 Warm-Up: Which One Doesn't Belong?

Four inequalities are shown.

Inequality A	Inequality B	Inequality C	Inequality D
$x \geq 5$	$12 \leq 7$	$2 > x$	$x > 1$

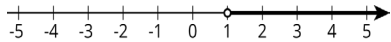
- Javier says that Inequality B does not belong with the others. What is one reason Javier might have said this?
The inequality B is wrong because 12 is greater than 7.
- Select a different inequality and explain why it doesn't belong.
Answers will vary.
 $2 > x$
I chose this one because in all the other equations x is greater.



10.1.2 Refresher: Representing Inequalities on a Number Line

Work with your partner to complete the following.

The values that make the inequality $x > 1$ true are represented on the number line shown.



1. Select the values of x from this list that could make the inequality $x > 1$ true.

- ☐ $3\frac{1}{8}$
- ☐ $-3\frac{7}{8}$
- ☐ 1
- ☐ 700
- ☐ 1.05
- ☐ -16.4

The x -values that make this inequality true are called solutions; however, there are many more possible solutions than those represented on the list.

2. Name two more values of x that are solutions to the inequality.
Answers will vary. $x = 3$, $x = 4$
3. Explain how you know the values are solutions.
I know the values are solutions because the values are greater than 1.
4. Name two values of x that are **not** solutions to the inequality.
Answers will vary. $x = -2$, $x = -3$
5. Explain how you know these values are **not** solutions.
I know the values are not solutions because the values are less than 1.
6. Alexis says the graph on the number line also represents the solutions to the inequality $1 < x$. Discuss with your partner whether you agree with Alexis, then write a brief summary of your discussion.
Yes, because it is still representing all values greater than 1.

10.1.2 REFRESHER: REPRESENTING INEQUALITIES ON A NUMBER LINE

TEACHER MOVES

Determine if your students need the support of all or part of the refresher.

ENRICHMENT

What does the open circle at 1 mean? Is 1 a solution to the inequality $x > 1$?

TEACHER MOVES

Students may struggle understanding that inequalities of the form $x > a$ and a



10.1.3 Exploration: Representing Inequalities in Context

Height restrictions are often in place for roller coasters and rides at amusement parks to keep guests safe. At Disney's Typhoon Lagoon Water Park, the following height restrictions are in place.

Attraction	Height Restriction
Bay Slides	Guests must be 60 inches or shorter
Crush 'n' Gusher	Guests must be 48 inches or taller
Ketchakiddee Creek	Guests must be 48 inches or shorter

1. Work with your partner to determine which graph represents the height restrictions of each attraction at Typhoon Lagoon. Then, draw an arrow from each attraction to its matching graph.

Attraction	Graph Representing Safe Heights for the Attraction
Bay Slides	
Crush 'n' Gusher	
Ketchakiddee Creek	

2. Nick is happy to realize that he is tall enough to go on the Crush 'n' Gusher raft ride. Nick's friend Matt is 2 inches shorter than Nick. Explain whether Matt is tall enough to go on the ride with Nick.
Answers will vary. If Nick is 48 inches tall then Matt would not be tall enough because he would be 46 inches. However, if Nick is 50 inches or taller then Matt is tall enough because he would be at least 48 inches.

10.1.3 EXPLORATION: REPRESENTING INEQUALITIES IN CONTEXT

ENRICHMENT

- How do you determine from the height restriction chart whether the circle is closed or open?
- How do you determine from the height restriction chart which direction to shade?
- What values make sense in this context?

10.1.3 EXPLORATION: REPRESENTING INEQUALITIES IN CONTEXT (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.

DIFFERENTIATE INSTRUCTION

For question 3, consider modeling Guess and Test for students. For example, “If Nick is 52 inches tall, then how tall is Matt?” Students should respond with 50 inches. This example will help students understand why the expression $n-2$ represents the context.

QUESTIONING

Question 6 is a good place to offer an extension. Ask students how they think they could solve the inequality using algebraic thinking.

3. Complete the statements.

If Nick is n inches tall, then Matt is $\begin{matrix} \circ n+2 \\ \circ n-2 \end{matrix}$ inches tall.

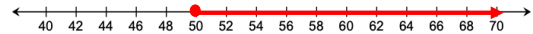
For Matt to be tall enough to go on Crush 'n' Gusher,

the value of $\begin{matrix} \circ n+2 \\ \circ n-2 \end{matrix}$ must be $\begin{matrix} \circ \text{less than} \\ \circ \text{greater than} \end{matrix}$ or equal to $\begin{matrix} \circ 48. \\ \circ 60. \end{matrix}$ Therefore, the solutions to the

inequality $\begin{matrix} \circ n+2 \leq \\ \circ n+2 \geq \\ \circ n-2 \leq \\ \circ n-2 \geq \end{matrix}$ $\begin{matrix} \circ 48 \\ \circ 60 \end{matrix}$ represent Nick's height

that mean Matt could also ride the Crush 'n' Gusher.

4. On the number line, show all the possible heights that Nick could be if Matt is also tall enough to go on the ride.



5. What is the minimum height Nick must be in order for Matt to be able to go on the ride as well?

Nick must be at least 50 inches for Matt to go on the ride as well.

Julia noticed the graph also represents the solutions to the inequality $n \geq 50$.

6. Work with your partner to explain why the solutions to the inequality $n-2 \geq 48$ can be represented the same way on a number line as $n \geq 50$.

Answers will vary. The solution to $n-2 \geq 48$ is the same set of numbers as the solution to $n \geq 50$, therefore the graphs will be the same.



10.1.4 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Inequalities have an infinite number of solutions. The set of all the solutions of an inequality is called the **solution set**.

The **solution set** of an inequality includes the values that, when substituted for the variable, make the inequality true.

One way to represent the solution set of an inequality is by shading the values on the number line, beginning with the minimum or maximum value, and including all the values to the right or left of it that are included in the solution set.

1. Complete each statement.

A minimum or maximum value represented by an open circle means the value is
☐ included in
☒ excluded from
 the solution set.

A minimum or maximum value represented by a closed circle means the value is
☒ included in
☐ excluded from
 the solution set.

Graphs representing the solution set of inequalities are helpful when interpreting them in real-world contexts. However, sometimes there are parts of the solution set that may not make sense in a given context.

For example, to ride the Bay Slides at Typhoon Lagoon, guests must be 60 inches or shorter. The graph below represents the graph of $h \leq 60$ in this situation, where h is the height of guests that can ride the Bay Slides.



2. What does 60 represent in this situation?
60 represents the maximum height of guests that can go on Bay Slides.
3. What does the arrow pointing to the left on the number line represent?
The arrow pointing towards left represent the heights of guests than can ride on Bay Slides.
4. Describe at least two values that are part of the solution set of $h \leq 60$ that do not make sense in this context.

Answers will vary. $h = -2$ or $h = -0.25$ because you cannot have negative heights.



Sometimes there are values that are part of the mathematical solution set that do not make sense in the context. In those cases, the variable may have a **constraint** (restriction or limitation) placed on it so that the mathematical model is realistic.

The Exploration demonstrated the inequalities $n - 2 \geq 48$ and $n \geq 50$ are equivalent because all of the same values of n make both inequalities true. Both inequalities were also represented using the same graph.

Inequalities that have the same solution set are called equivalent inequalities.

10.1.4 BRING IT TOGETHER: INEQUALITIES IN MATHEMATICAL AND REAL-WORLD CONTEXTS

TEACHER MOVES

Before moving, check for understanding regarding the definition of solution set. Ask students what “an infinite number of solutions” means.

ENRICHMENT

- What is the minimum height you would expect to see on a water slide?
- What heights are too low?
- Are negative solutions reasonable in the context of this question? Why or why not?

DIFFERENTIATE INSTRUCTION

Students may struggle comprehending the word “constraint.” Consider using a Vocabulary Improvement Strategy (V.I.S.) to connect the meaning of the word to other contexts and a visual representation of what it means to the student.

10.1.5 COLLABORATION: EXPLORING INEQUALITIES

TEACHER MOVES

Students should be using substitution here to determine which solutions will work and which ones will not. Encourage them to substitute values from the shaded portion of the graph to determine which graphs will represent which equations.

ENRICHMENT

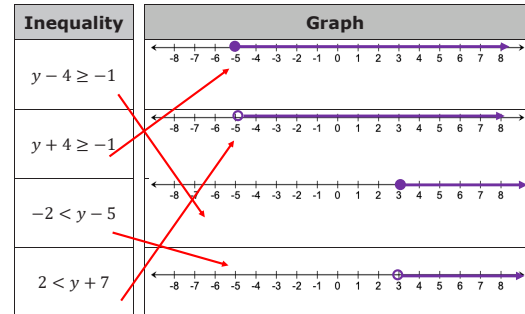
- Why is the minimum value negative infinity?
- Why is 1 not the maximum value?
- How can the value that is closest to 1 but is not 1 be represented?



10.1.5 Collaboration: Exploring Inequalities

Work with your partner to complete the following.

1. Match each inequality with the graph that represents all the values of y that make it true.



2. Choose one inequality and explain how you determined which graph represents its solutions.

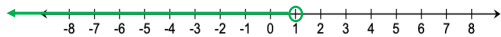
$2 < y + 7$

Answers will vary. I would first eliminate graphs with closed or open circles based on the sign of the inequality. Then, I would determine whether the graph will have a maximum or minimum based on the inequality. Lastly, if needed, I would substitute numbers into the inequality to see if they belong in the solution set of the graph.

3. What is the minimum value that makes the inequality $y - 4 \geq -1$ true?

$y = 3$

4. The graph of an inequality is shown.



Circle the inequalities that can be represented by the graph.

$m + 1.2 < 2.2$

$m - 1.2 < 2.2$

$\frac{1}{4} > m + \frac{3}{4}$

$1 > m$

$1 < m$

$\frac{1}{4} > m - \frac{3}{4}$

5. Complete the statement.

The inequalities circled above are equivalent inequalities.

6. Suppose the solution set shown on the number line represents the possible height of Krista's feet in relation to the surface of the water in her pool when she jumps off the pool's deck. Describe any possible constraints on the solution set in this context.

Answers will vary. Large negative numbers that are unrealistic for the depth of the pool would not make sense in this context. (i.e., -65, which represents 65 feet below the surface of the water)

7. Describe the minimum or maximum value that makes this inequality true given the context.

Answers will vary. The maximum value is a value that is close to 1 but not equal to 1. This value could be represented by 0.9.

10.1.5 COLLABORATION: EXPLORING INEQUALITIES (CONTINUED)

ENRICHMENT

- If one number on the shaded portion of the graph is not true for the inequality, does that mean the whole graph is not true?
- What number would you choose to substitute into the inequality to check your answer? Why?

DIFFERENTIATE INSTRUCTION

Consider providing students with a collection of numbers to substitute in the inequalities with Guess and Check. This will help students match the inequalities with the number lines and determine the maximum or minimum values.



10.1.6 Wrap-Up: Cool Down

1. Complete the table.

Answers will vary.

The terms and information used in today's lesson that I knew were...	Something I wondered during today's lesson was...	Something I learned during today's lesson was...

10.1.6 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

For students who struggle representing their conceptual understanding through writing, consider providing alternative ways to assess learning for this lesson. Students could use an audio recorder or speech-to-text software, draw and label a diagram, or provide examples. All forms show an equal level of rigor and conceptual understanding.